

host-parasite coevolution



RESISTANCE vs TOLERANCE

(close compatibility filter)

(reducing the loss of fitness)

(*)

- RED QUEEN dynamics.

new alleles arise frequently

frequency-dependent selection

rare alleles have an advantage

genetic systems

[self-nonself recognition system]

= parasite can infect + persist (compatibility filter OPEN)

if it matches a corresponding parasitism host allele

	I	II	III
1	✓		
2		✓	
3			✓

✓ = parasite wins
MATCHING ALLELES

	I	II	III
1		✓	✓
2	✓		✓
3	✓	✓	

INVERSE matching allele
(antibodies/adaptive immunity)

	I	II	III
1	✓		
2	✓		✓
3	✓	✓	✓

GENE-FOR-GENE
resistance gene in hosts

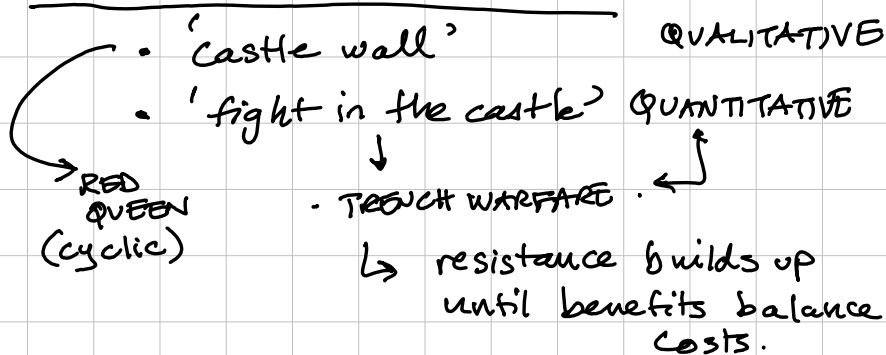
(a) virulence gene in parasites

Some of these patterns lead to cycles.

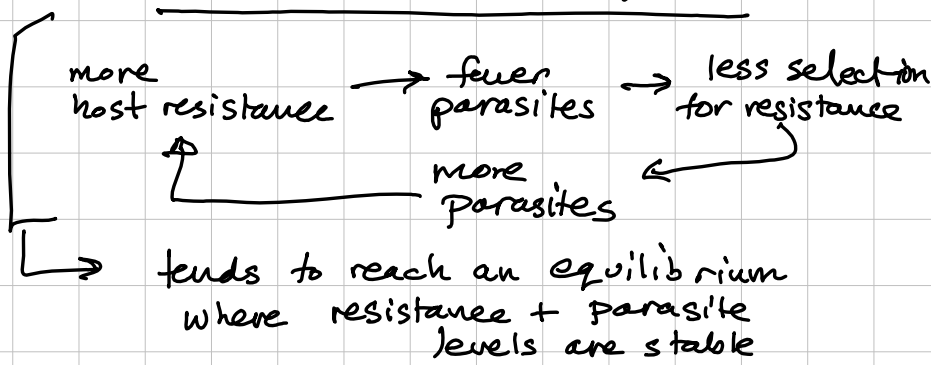
inverse matching

parasites: I → II → III

hosts: 1 → 2 → 3



$$\text{energetic cost} = \left(\text{prob of parasitism} \right) \times \text{costs of parasitism}$$



resistance alleles are POLYMORPHIC.

DIOECY
(DIOECIOUS)
(vs monoecious)

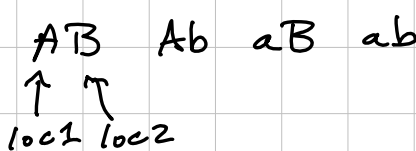
GONOCORY

harmaphrodites
simultaneous or sequential
SELFING or OUTCROSSING

COSTS OF SEX :

mating failure
cost of meiosis &
cost of OUTBREEDING (EPISTASIS)

(AN)ISO GAMY



advantages of sex